project.org

STUDENT HANDOUT

RED LEVEL
GO DEEPER

Math vs. Your Eyes

Grades 5–8 · Math

ADMIT ONE

NAME _____ DATE _____ CLASS _____

The geometry of deception. Mathematicians and artists have studied optical illusions for centuries. The principles behind these illusions—perspective, proportion, and angular relationships—are the same ones used in architecture, engineering, and computer graphics.

01 PART 1 Quantifying the illusion

The Müller-Lyer illusion's strength varies with arrow angle. Test this mathematically:

Draw 3 versions of the illusion with arrows at 30°, 45°, and 60°. Survey 3 people: *"How much longer does line A appear than line B?"*

ARROW ANGLE	PERSON 1 ESTIMATE	PERSON 2 ESTIMATE	PERSON 3 ESTIMATE	AVERAGE
30°				
45°				
60°				

Conclusion: At what angle is the illusion strongest? _____°

Why might this be? (Think about room corners and depth perception)

02 PART 2 Design a forced perspective illusion

Forced perspective uses mathematical scaling to make objects appear larger or smaller than they are. Famous examples: The Leaning Tower of Pisa tourist photos, movie sets (hobbits in *The Lord of the Rings*).

Design your own forced perspective scene. Include:

- A sketch showing the camera position and object placement
- Calculations showing the scale ratios needed
- The mathematical relationship between distance and apparent size

Key formula: Apparent size is inversely proportional to distance.

If object A is 2× farther than object B, it appears _____ the size.

If object A is 3× farther than object B, it appears _____ the size.

03

PART 3

Mathematical explanation

Write a mathematical explanation of why the Ponzo illusion works. Use these terms: *converging lines, ratio, proportion, perceived distance*

04

PART 4

Forced perspective design

Sketch your forced perspective scene below. Label distances and show your calculations:

DRAW YOUR SCENE HERE

My calculations:

Distance from camera to near object: _____ m Distance from camera to far object: _____ m

Ratio: _____ : _____

To make the far object appear the SAME size as the near object, it needs to actually be _____ times larger.

MINI MUSEUM CONNECTION

Mini Museum deep dive: The stereoscope creates depth through **parallax**—the angular difference between two views.

Research question: How do surveyors use parallax to calculate distances? What's the mathematical relationship between baseline distance, parallax angle, and perceived depth?

Visit: *minimuseumproject.org/exhibitions/seeing-is-deceiving/illusions*

05 PART 5

Advanced analysis

The human eye has a field of view of approximately 120°. A standard camera has about 45°.

Question: How might this difference affect how optical illusions appear in photographs versus real life?

06 PART 6

Extension: anamorphic art

Anamorphic art uses extreme perspective distortion—images that only look correct from one specific angle. Research one example (like Julian Beever's pavement art or Hans Holbein's *The Ambassadors*).

What mathematical transformation is being applied?

SHARE WITH YOUR FAMILY

Project: Create a forced perspective photo with family members. Calculate the exact positioning needed to make someone appear to be "holding" a distant building or "stepping on" a family member. Show your math!